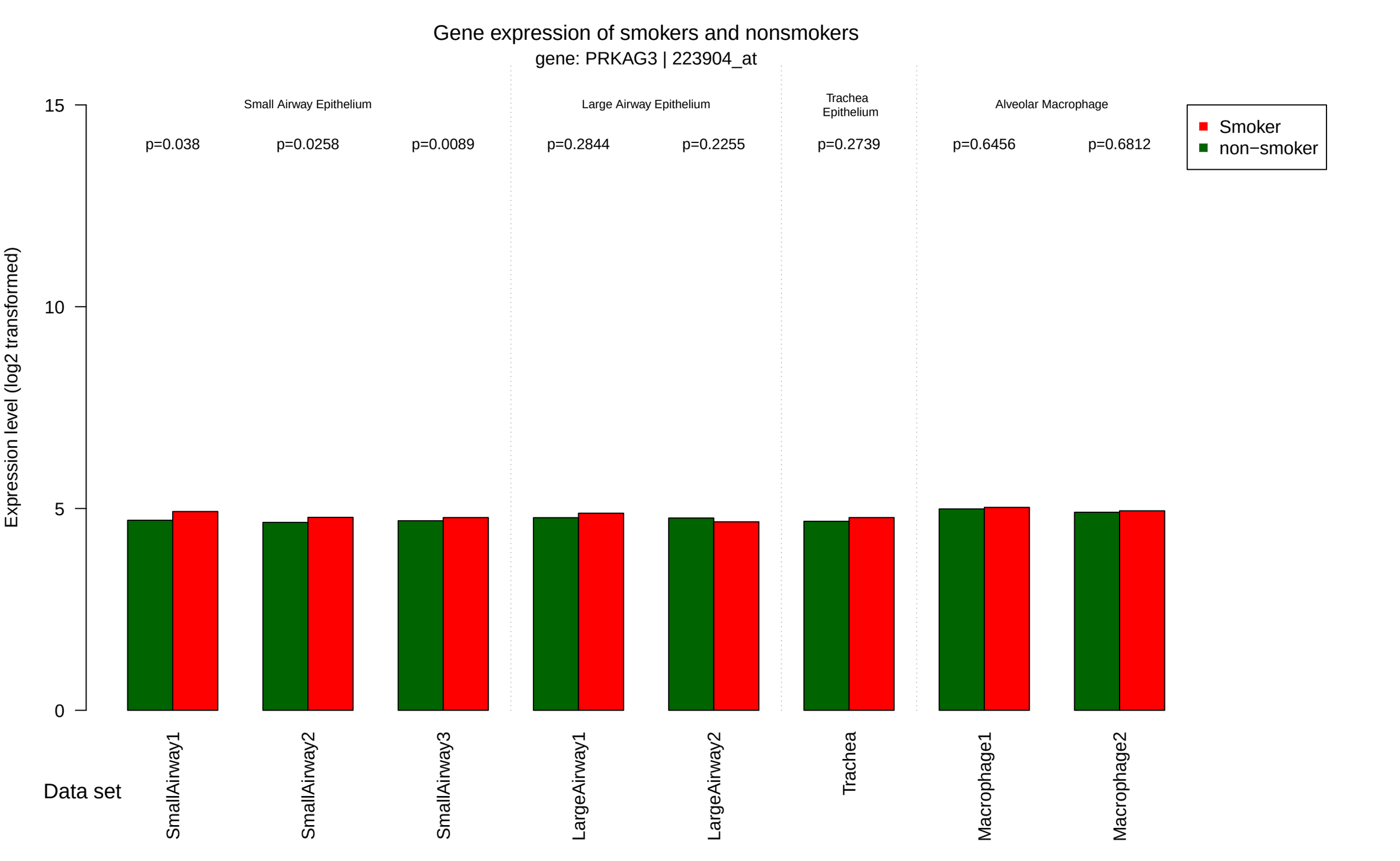
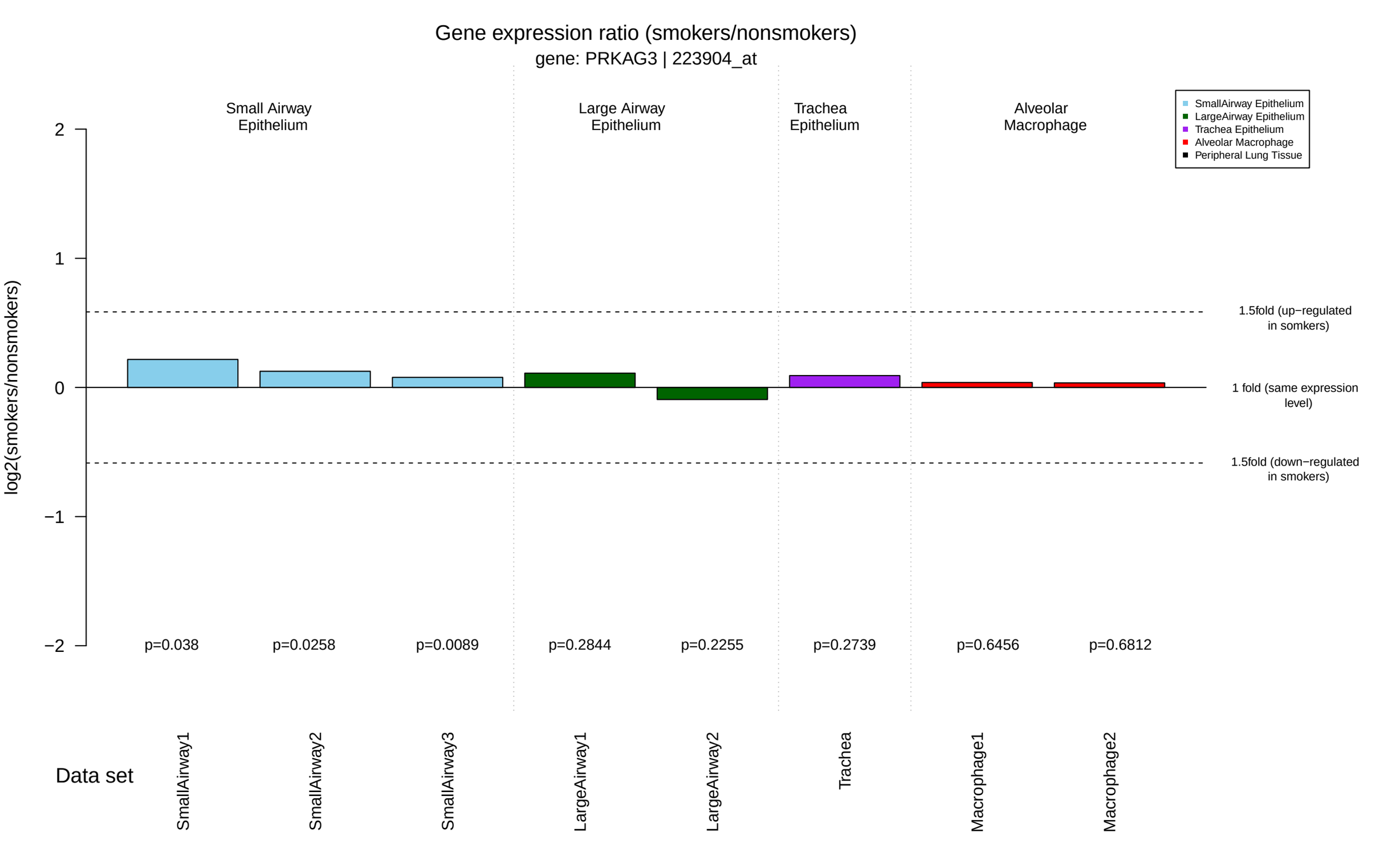
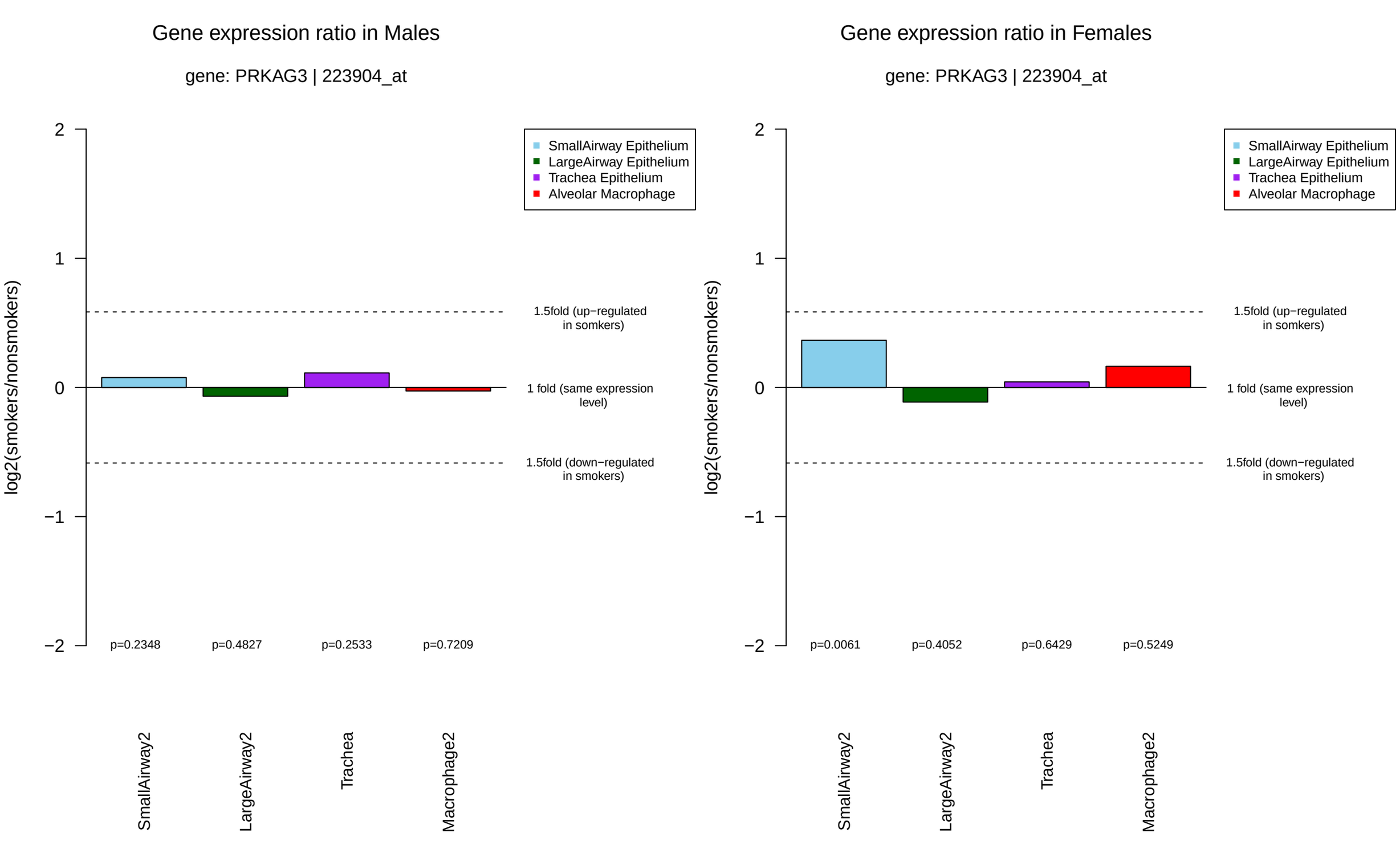
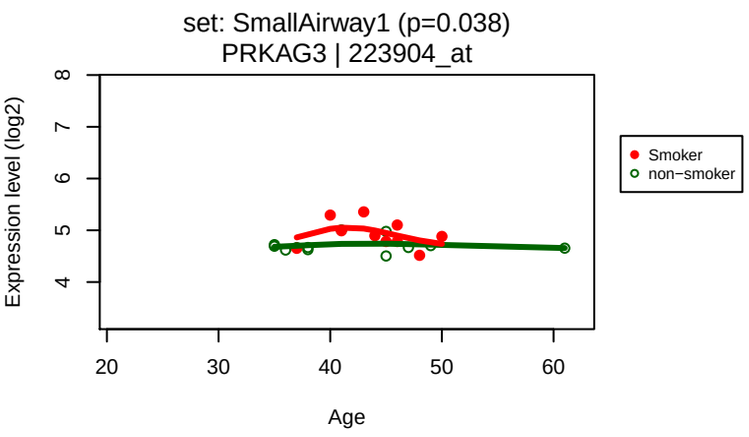
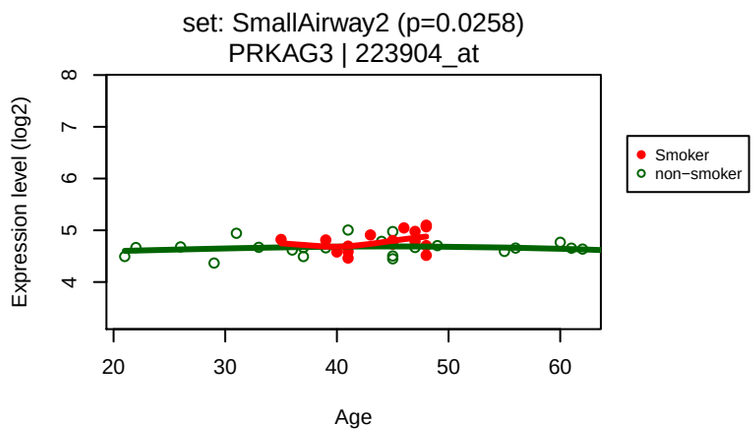
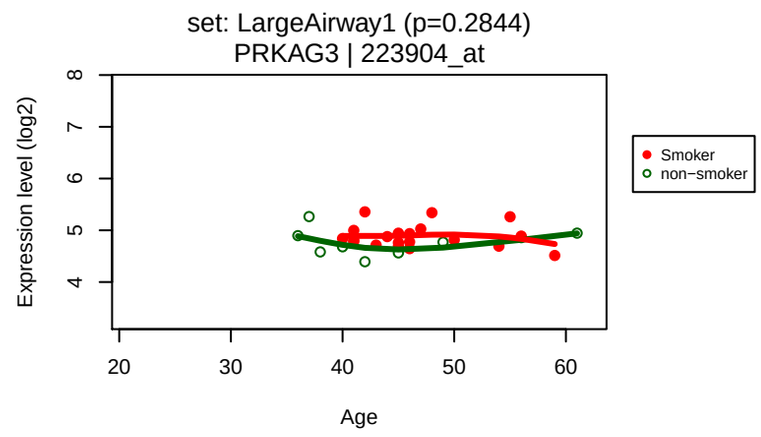

Gene expression of smokers and nonsmokers
gene: PRKAG3 | 223904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.038
p=0.0258
p=0.0089
p=0.2844
p=0.2255
p=0.2739
p=0.6456
p=0.6812
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

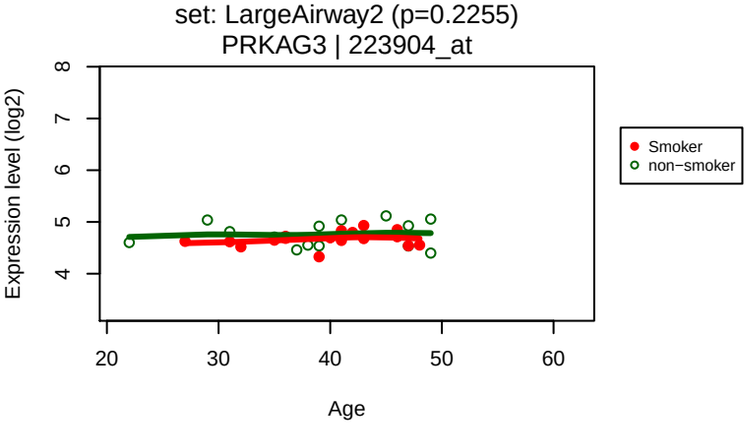

Gene expression ratio (smokers/nonsmokers)
gene: PRKAG3 | 223904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.038
p=0.0258
p=0.0089
p=0.2844
p=0.2255
p=0.2739
p=0.6456
p=0.6812
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

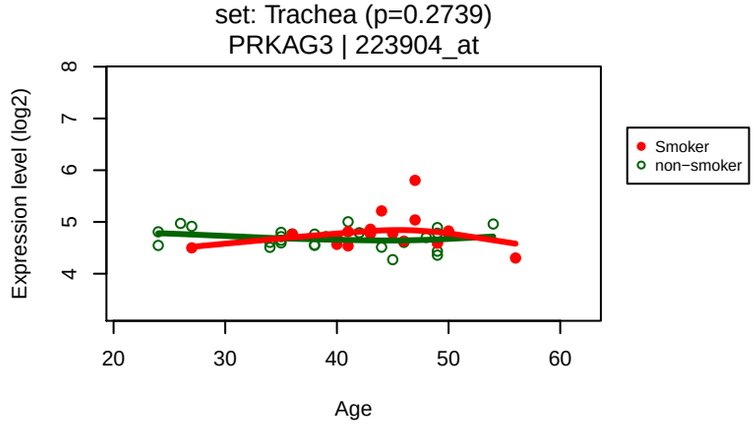

Gene expression ratio in Males
gene: PRKAG3 | 223904_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2348
p=0.4827
p=0.2533
p=0.7209
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAG3 | 223904_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0061
p=0.4052
p=0.6429
p=0.5249
SmallAirway2
LargeAirway2
Trachea
Macrophage2

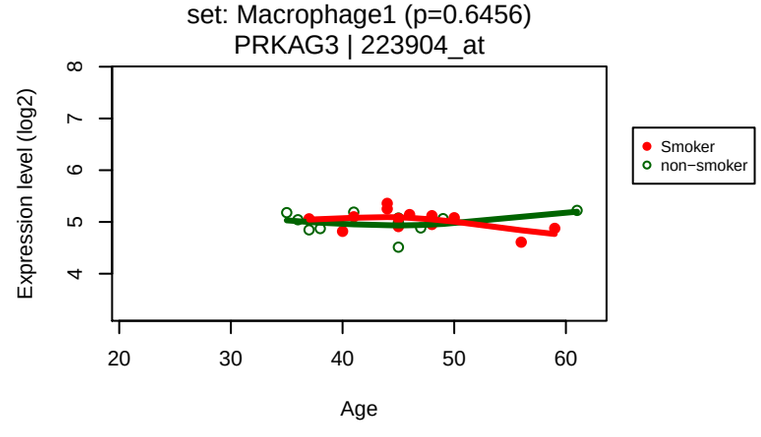

set: SmallAirway1 (p=0.038)
PRKAG3 | 223904_at
Expression level (log2)
Age
Smoker
non−smoker

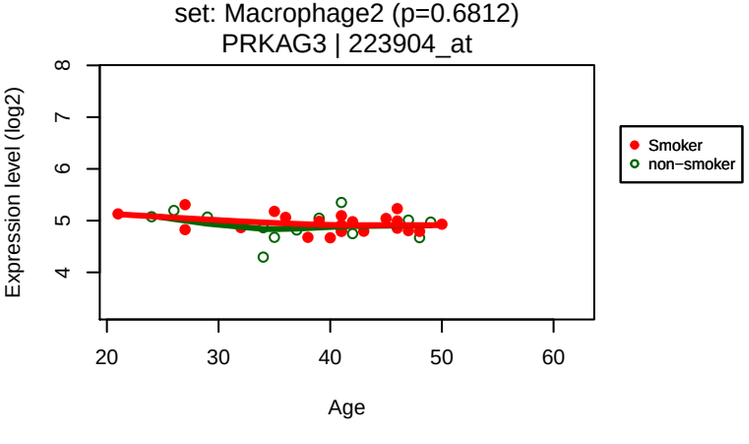

set: SmallAirway2 (p=0.0258)
PRKAG3 | 223904_at
Expression level (log2)
Age
Smoker
non−smoker

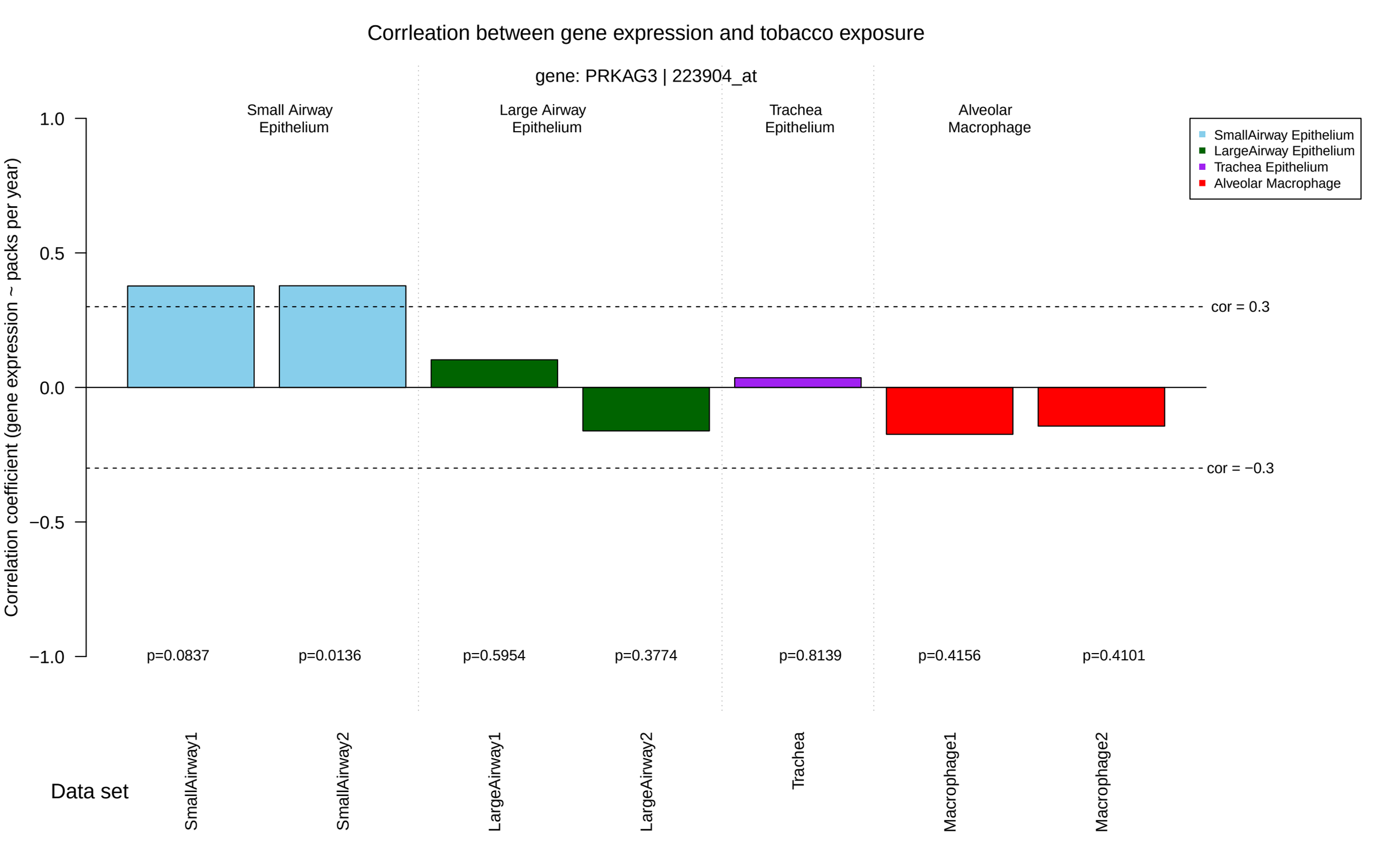

set: LargeAirway1 (p=0.2844)
PRKAG3 | 223904_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2255)
PRKAG3 | 223904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2739)
PRKAG3 | 223904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6456)
PRKAG3 | 223904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6812)
PRKAG3 | 223904_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAG3 | 223904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0837
p=0.0136
p=0.5954
p=0.3774
p=0.8139
p=0.4156
p=0.4101
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2